



Using Logarithms to Solve Exponential Equations

To solve an exponential equation such as $2^x = 10$, you have to use a logarithm.

Just as you use addition, multiplication, or square roots to transform and solve equations, you can use logarithms to transform and solve equations.

Using the Definition of Logarithms to Solve Exponential Equations

EXAMPLE 1

Solve. Round your answers to four decimal places if necessary.

A $3^x = 81$

SOLUTION

If you raise the base, 3, to the fourth power, you get 81.

$$3^x = 81$$

$$3^x = 3^4$$

$$x = 4$$

B $e^x = 10$

SOLUTION

There is no whole number you can raise e to and get 10 (or any other whole number greater than 1). Use the definition of the logarithm.

$$\begin{aligned} e^x = 10 &\Rightarrow x = \log_e 10 \\ &x = \ln(10) \approx 2.3026 \quad \blacksquare \end{aligned}$$

Using Logarithmic Transformations to Solve Exponential Equations

Property of Equality for Logarithmic Equations

If a , x , and y are positive numbers and $a \neq 1$, then $\log_a x = \log_a y$ if and only if $x = y$.

EXAMPLE 2

Solve. Round your answer to four decimal places.

► **Think About It** It is customary to round the value of a logarithmic expression to four decimal places.

A $2^x = 10$

SOLUTION

$$2^x = 10$$

$$\log 2^x = \log 10 \quad \text{Take the common (base 10) logarithm of both sides.}$$

$$x \log 2 = \log 10 \quad \text{Power Property of Logarithms}$$

$$x \log 2 = 1 \quad \log_{10} 10 = 1 \text{ because } 10^1 = 10.$$

$$x = \frac{1}{\log 2} \quad \text{Divide both sides by } \log 2.$$

$$x \approx 3.3219 \quad \text{Use a calculator to approximate the expression.}$$

Check for reasonableness. Because $2^3 = 8$ and $2^4 = 16$, it is reasonable that $2^{3.3219} \approx 10$. You may use a calculator to verify the result.

B $5^{2n} = 101$

SOLUTION

$$5^{2n} = 101$$

$$\log 5^{2n} = \log 101$$
 Take the common logarithm of both sides.

$$2n \log 5 = \log 101$$
 Power Property of Logarithms

$$n = \frac{\log 101}{2 \log 5}$$
 Divide both sides by $2 \log 5$.

$$n \approx 1.4338$$
 Use a calculator to approximate the expression. ■

► **Think About It** Always solve the equation thoroughly before approximating the value of the variable. The equation $n = \frac{\log 101}{2 \log 5}$ gives the exact value of n ; $n \approx 1.4338$ gives an approximation.

When an exponential equation contains the number e , it can be easier to take the natural logarithm of both sides to solve the equation.

► **Think About It** The equations in Example 2 could also have been solved using the natural logarithm. For an equation with a variable in the exponent, you can choose to use either the natural or the common logarithm.

EXAMPLE 3

Solve the equation by using logarithms. Round your answer to four decimal places.

A $e^{3a} = 9$

SOLUTION

$$e^{3a} = 9$$

$$\ln e^{3a} = \ln 9 \quad \text{Take the natural logarithm of both sides.}$$

$$3a \ln e = \ln 9 \quad \text{Power Property of Logarithms}$$

$$3a = \ln 9 \quad \ln e = 1 \text{ because } e^1 = e.$$

$$a = \frac{\ln 9}{3} \quad \text{Divide both sides by 3.}$$

$$a \approx 0.7324 \quad \text{Use a calculator to approximate the expression.}$$

CHECK

To verify the result, substitute the approximate value in the original equation. Use a calculator to evaluate the left side of the equation.

$$\begin{aligned} e^{3a} &= 9 \\ e^{3 \cdot 0.7324} &\approx 9 \\ 8.9998 &\approx 9 \quad \checkmark \end{aligned}$$

The solution $a \approx 0.7324$ is correct.

B $7^x = e^{x+1}$

Method Using LN

$$\begin{aligned} 7^x &= e^{x+1} \\ \ln 7^x &= \ln e^{x+1} \\ x \ln 7 &= (x+1) \ln e \\ x \ln 7 &= (x+1) \cdot 1 \\ x \ln 7 &= x+1 \\ x \ln 7 - x &= 1 \\ x((\ln 7) - 1) &= 1 \\ x &= \frac{1}{(\ln 7) - 1} \\ x &\approx 1.0572 \end{aligned}$$

Method Using LOG

$$\begin{aligned} 7^x &= e^{x+1} \\ \log 7^x &= \log e^{x+1} \\ x \log 7 &= (x+1) \log e \\ x \log 7 &= x \cdot \log e + 1 \cdot \log e \\ x \log 7 - x \log e &= \log e \\ x(\log 7 - \log e) &= \log e \\ x &= \frac{\log e}{\log 7 - \log e} \\ x &\approx 1.0572 \end{aligned}$$

CHECK

To verify the result, substitute the approximate value in the original equation. Use a calculator to evaluate each side of the equation.

$$\begin{aligned}7^x &= e^{x+1} \\ 7^{1.0572} &\approx e^{1.0572+1} \\ 7.8242 &\approx 7.8240 \quad \checkmark\end{aligned}$$

The calculator gives the same value, if rounded to three decimal places, for each expression. ■

Interpreting a Calculator Error Message

EXAMPLE 4

Solve $10^x + 2 = -6$.

SOLUTION

$$10^x + 2 = -6$$

$$10^x = -8 \quad \text{Subtract 2 from both sides.}$$

$$\log 10^x = \log(-8) \quad \text{Take the common logarithm of both sides.}$$

$$x \log 10 = \log(-8) \quad \text{Power Property of Logarithms}$$

$$x = \log(-8) \quad \log 10 = 1$$

When you try to use a calculator to evaluate $\log(-8)$, the calculator gives an error. There is no real solution to the equation. ■

► **Think About It** The statement $10^x = -8$ indicates that the original equation has no real solution. It is not possible for any power of 10 to be negative.